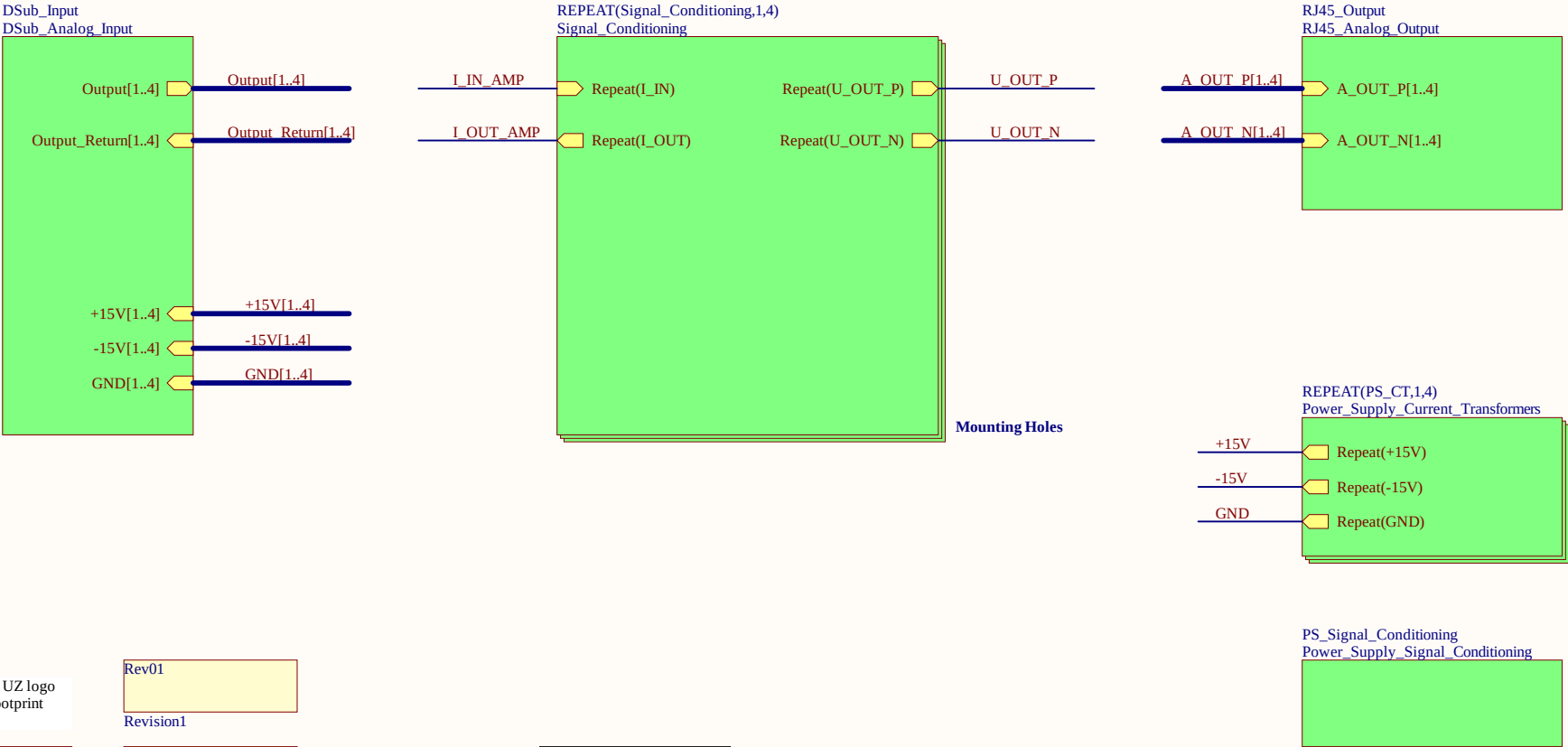


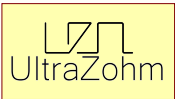
# TopSheet Adapter-Platine

|                |            |
|----------------|------------|
| Output1        | I_IN_AMP1  |
| Output_Return1 | I_OUT_AMP1 |
| Output2        | I_IN_AMP2  |
| Output_Return2 | I_OUT_AMP2 |
| Output3        | I_IN_AMP3  |
| Output_Return3 | I_OUT_AMP3 |
| Output4        | I_IN_AMP4  |
| Output_Return4 | I_OUT_AMP4 |

|          |          |
|----------|----------|
| U_OUT_P1 | A_OUT_P1 |
| U_OUT_N1 | A_OUT_N1 |
| U_OUT_P2 | A_OUT_P2 |
| U_OUT_N2 | A_OUT_N2 |
| U_OUT_P3 | A_OUT_P3 |
| U_OUT_N3 | A_OUT_N3 |
| U_OUT_P4 | A_OUT_P4 |
| U_OUT_N4 | A_OUT_N4 |



Chose size of UZ logo  
by chosing footprint  
type



LOGO1

Tom Gaupp

Designer1

Rev01

Revision1

uz\_per\_current\_measurement\_4channel  
Project1

04/26

Date1

Serial

Serial1

Place MH close to edge connector  
when ordering PCB (without  
assembly) from JLC. They ensure  
accurate outline milling of the  
edge connector.

MH1



MH2



MH3

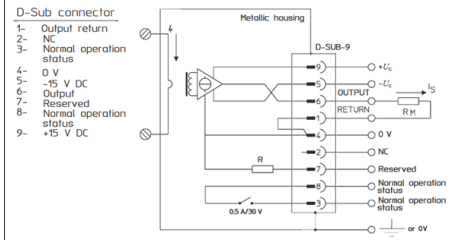


MH4



|                                                         |                            |                                                                                                                                                                                                                                                |
|---------------------------------------------------------|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Title TopSheet.SchDoc                                   |                            | <div>UltraZohm</div> <div><a href="http://www.ultrazohm.com">www.ultrazohm.com</a></div> <div>Date: 14/04/2026</div> <div>Sheet 1 of 12</div> <div></div> |
| Revision: Rev01                                         | Design Engineer: Tom Gaupp |                                                                                                                                                                                                                                                |
| Project: uz_per_current_measurement_dsub_to_rj45.PrjPcb |                            |                                                                                                                                                                                                                                                |

## Connection



# Signal Conditioning 1.4

Verstärkerschaltung Gain:  $G_{amp} = 2$

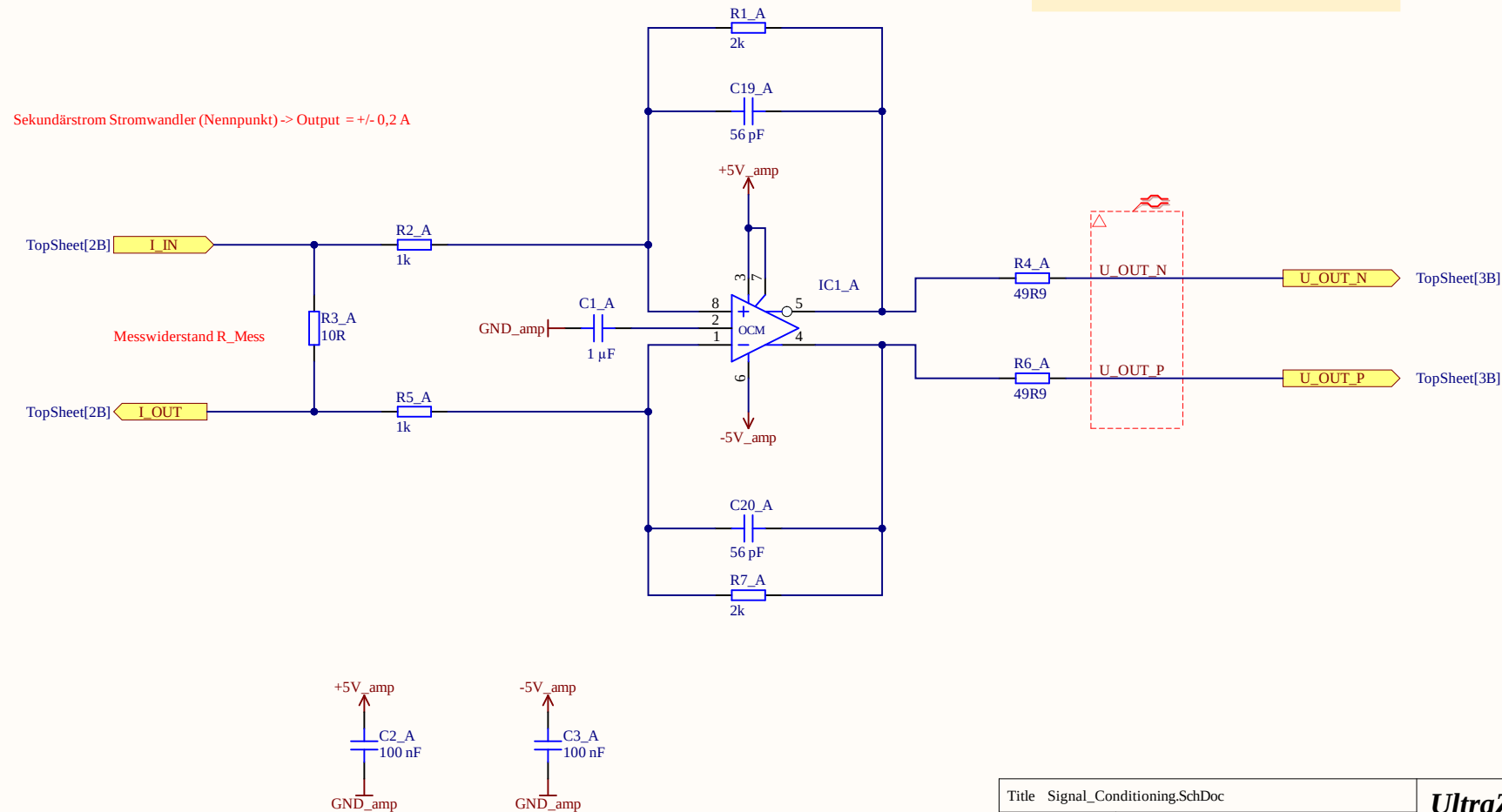
$$G_{amp} = R_F / R_G$$

Auslegung Rückkopplungskondensator für Tiefpassfilterung:

Grenzfrequenz:  $f_g$

Rückkopplungskondensator:  $C_f =$

$$C_f = 1 / 2 \cdot \pi \cdot R_f \cdot f_g = \dots =$$



Title Signal\_Conditioning.SchDoc

Revision: Rev01

Design Engineer: Tom Gaupp

Project: uz\_per\_current\_measurement\_dsub\_to\_rj45.PrjPcb

**UltraZohm**

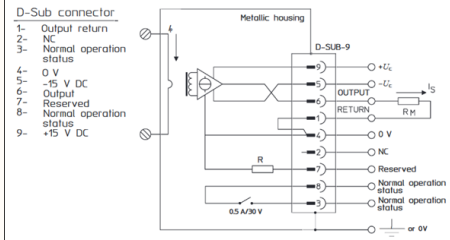
[www.ultrazohm.com](http://www.ultrazohm.com)

Date: 14/04/2026

Sheet 2.1 of 12



## Connection



# Signal Conditioning 1..4

Verstärkerschaltung Gain:  $G_{amp} = 2$

$G_{amp} = R_F / R_G$

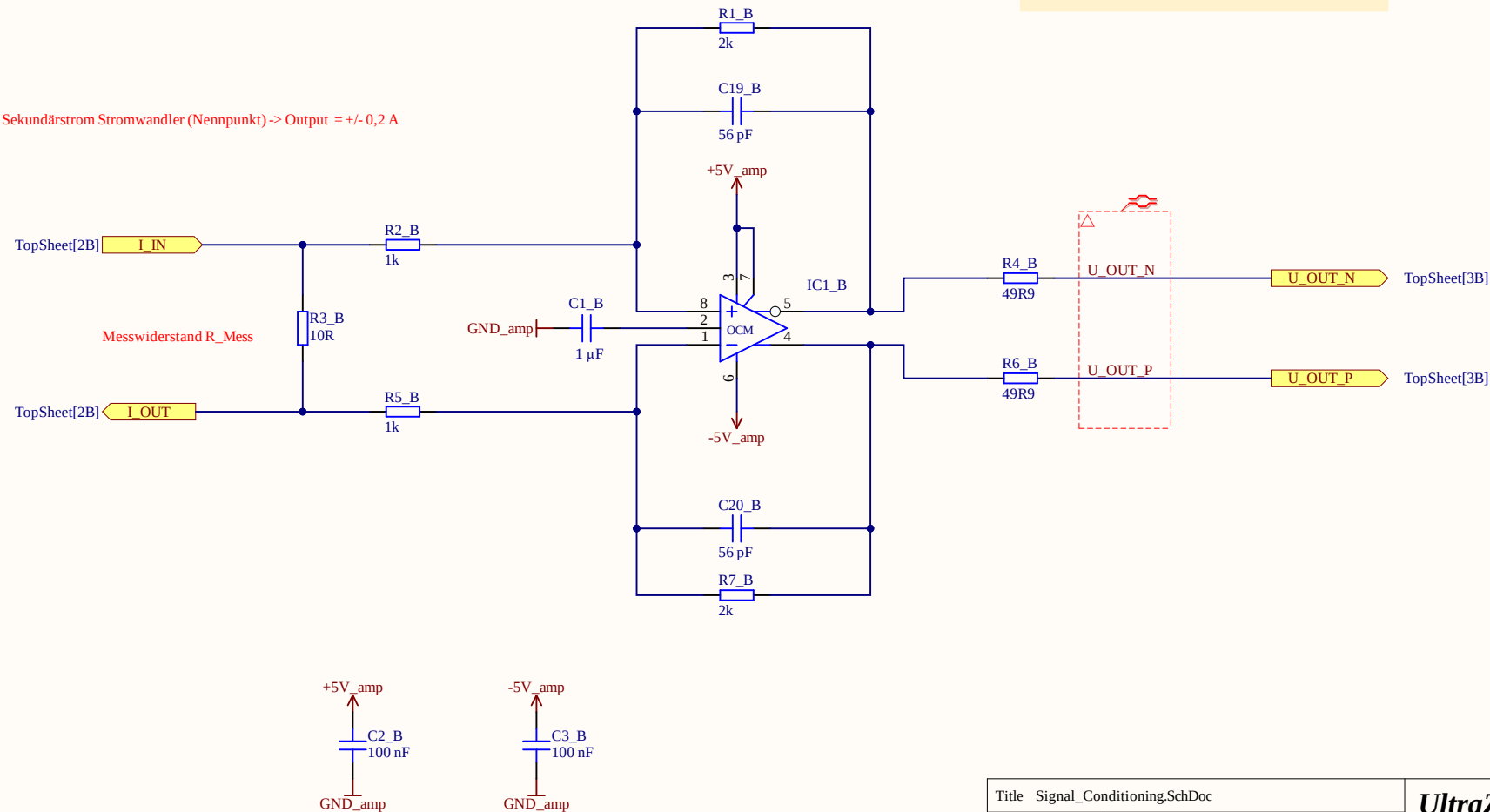
Auslegung Rückkopplungskondensator für Tiefpassfilterung:

Grenzfrequenz:  $f_g$

Rückkopplungskondensator:  $C_f =$

$C_f = 1 / 2 \cdot \pi \cdot R_f \cdot f_g = \dots =$

Sekundärstrom Stromwandler (Nennpunkt)  $\rightarrow$  Output =  $\pm 0,2$  A



Title: Signal\_Conditioning.SchDoc

Revision: Rev01

Design Engineer: Tom Gaupp

Project: uz\_per\_current\_measurement\_dsub\_to\_rj45.PrjPcb

**UltraZohm**

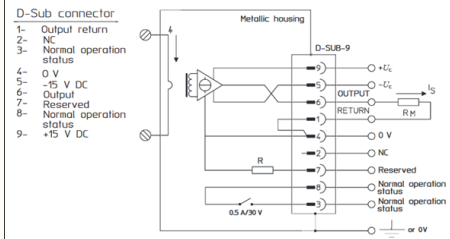
[www.ultrazohm.com](http://www.ultrazohm.com)

Date: 14/04/2026

Sheet 2.2 of 12



## Connection



# Signal Conditioning 1.4

Verstärkerschaltung Gain:  $G_{amp} = 2$

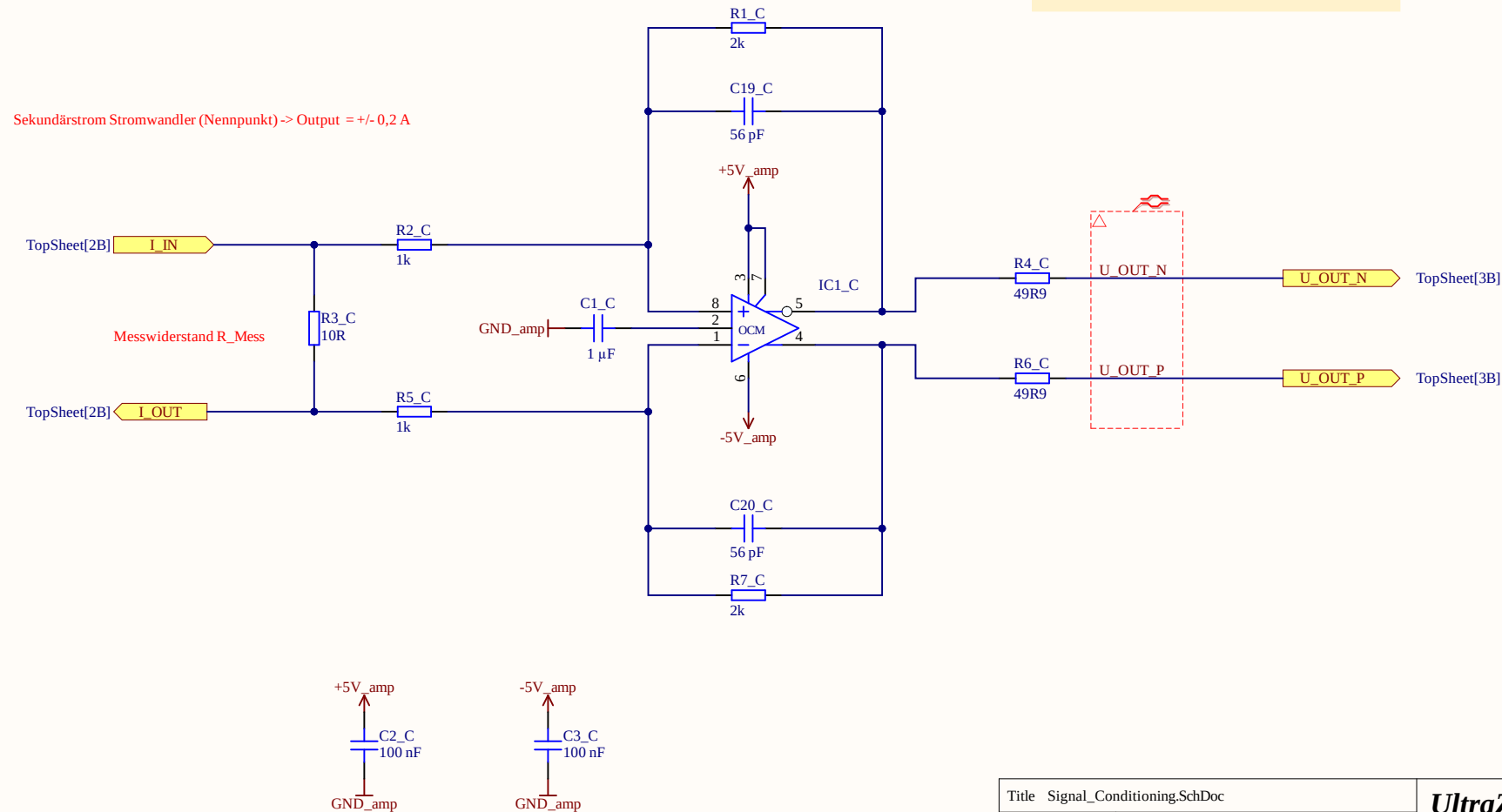
$G_{amp} = R_F / R_G$

Auslegung Rückkopplungskondensator für Tiefpassfilterung:

Grenzfrequenz:  $f_g$

Rückkopplungskondensator:  $C_f =$

$C_f = 1 / 2 \cdot \pi \cdot R_f \cdot f_g = \dots =$



Title: Signal\_Conditioning.SchDoc

Revision: Rev01

Design Engineer: Tom Gaupp

Project: uz\_per\_current\_measurement\_dsub\_to\_rj45.PrjPcb

**UltraZohm**

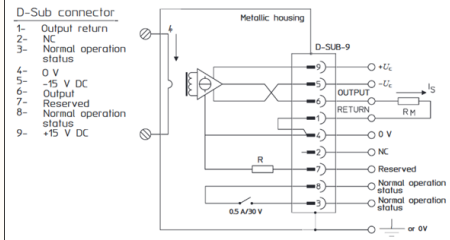
[www.ultrazohm.com](http://www.ultrazohm.com)

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## Connection



# Signal Conditioning 1.4

Verstärkerschaltung Gain:  $G_{amp} = 2$

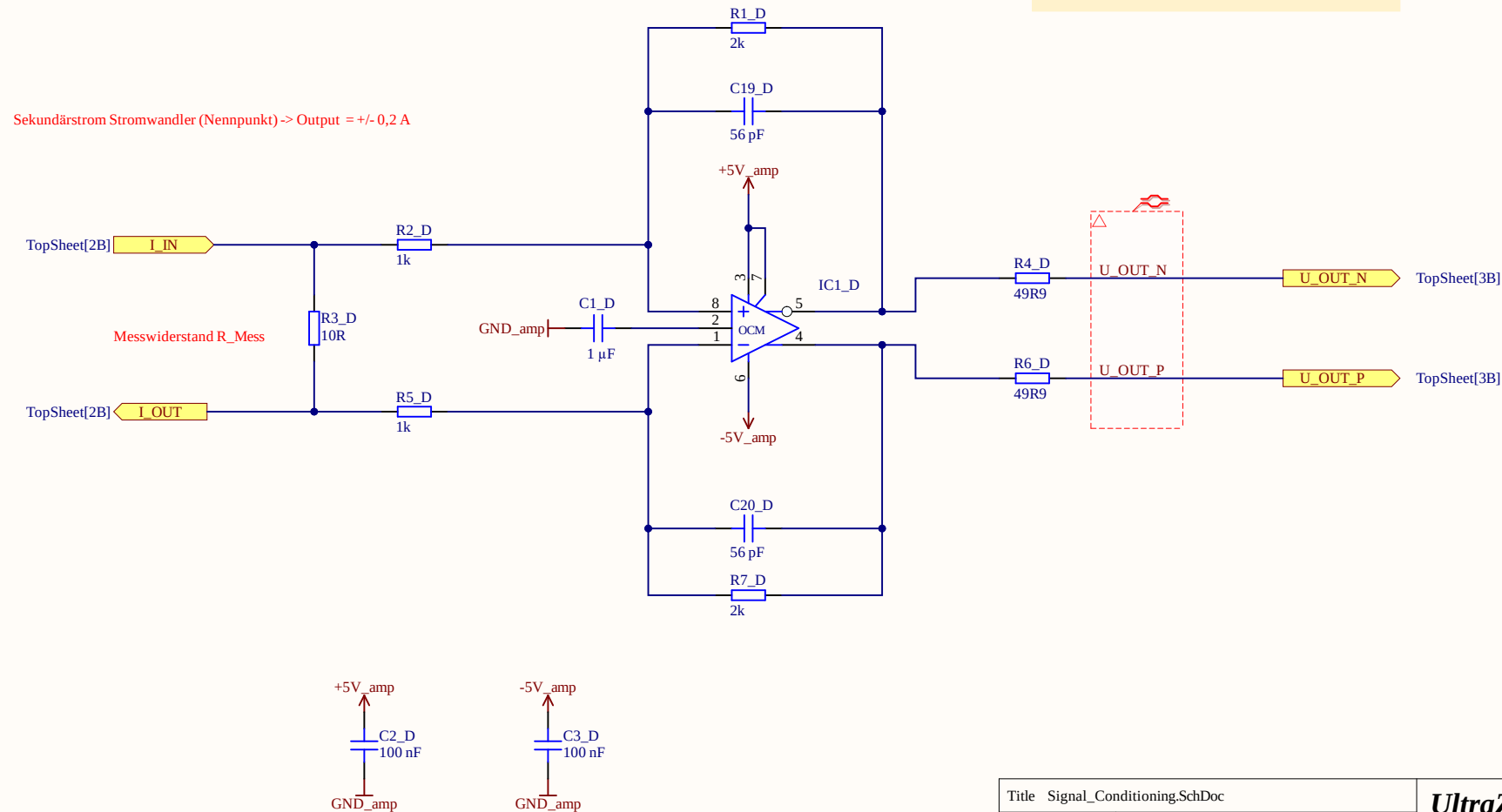
$G_{amp} = R_F / R_G$

Auslegung Rückkopplungskondensator für Tiefpassfilterung:

Grenzfrequenz:  $f_g$

Rückkopplungskondensator:  $C_f =$

$C_f = 1 / 2 \cdot \pi \cdot R_f \cdot f_g = \dots =$



Title Signal\_Conditioning.SchDoc

Revision: Rev01

Design Engineer: Tom Gaupp

Project: uz\_per\_current\_measurement\_dsub\_to\_rj45.PrjPcb

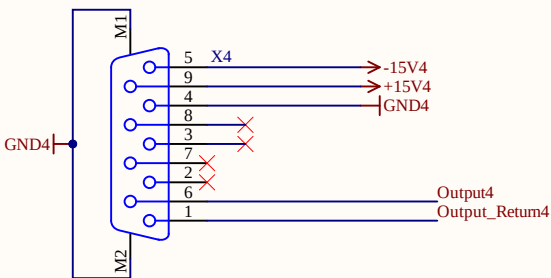
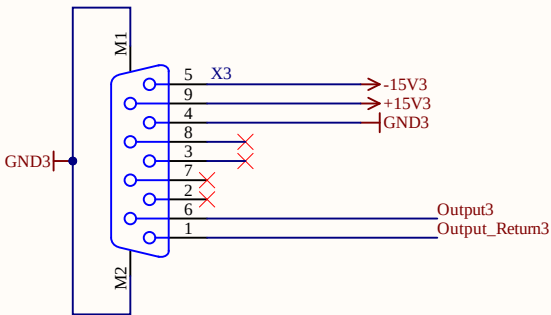
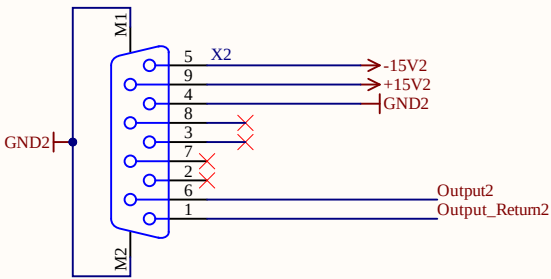
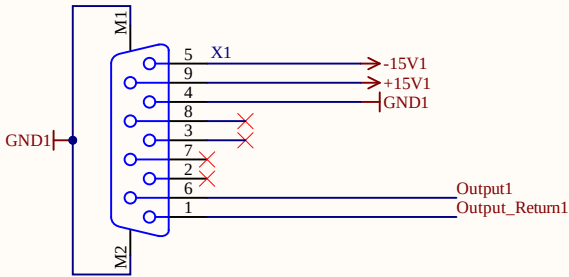
**UltraZohm**

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Date: 14/04/2026

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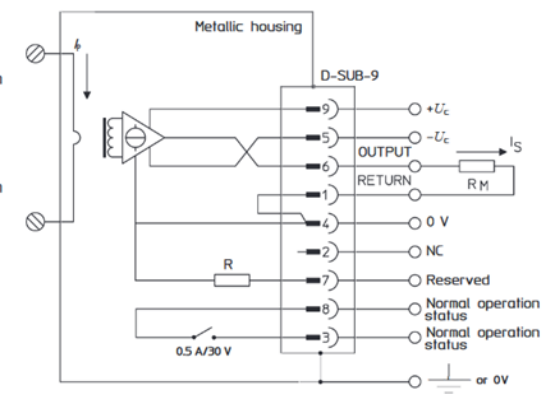




### Connection

#### D-Sub connector

- 1- Output return
- 2- NC
- 3- Normal operation status
- 4- 0 V
- 5- -15 V DC
- 6- Output
- 7- Reserved
- 8- Normal operation status
- 9- +15 V DC



Output[1..4] → Output[1..4] TopSheet[1B]

Output\_Return[1..4] → Output\_Return[1..4] TopSheet[1B]

+15V[1..4] → +15V[1..4] TopSheet[1B]

-15V[1..4] → -15V[1..4] TopSheet[1C]

GND[1..4] → GND[1..4] TopSheet[1C]

A

A

B

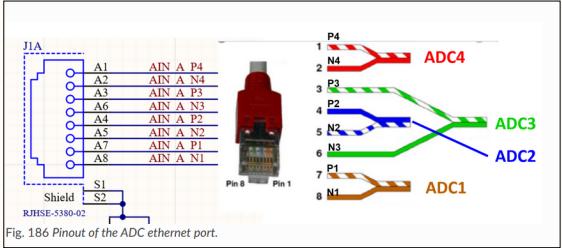
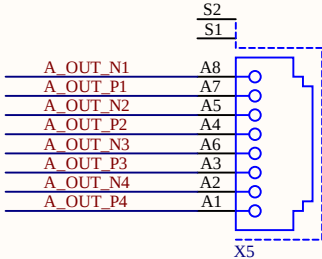
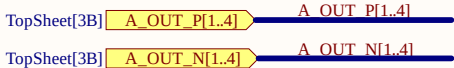
B

C

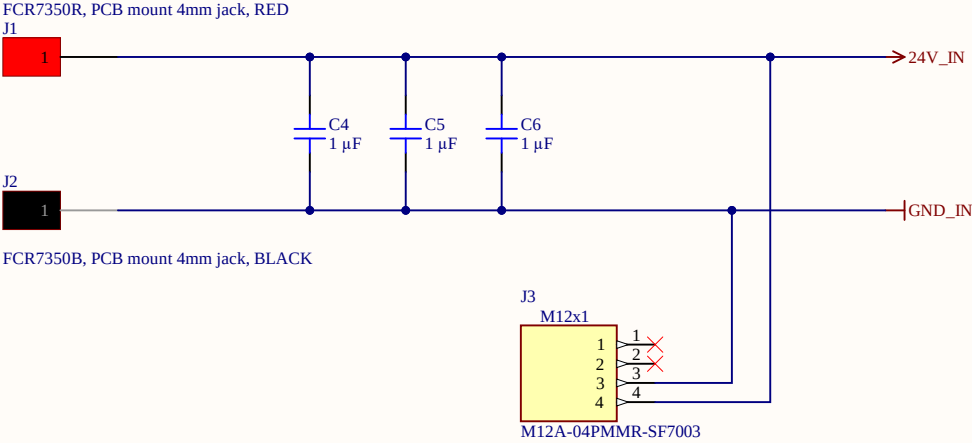
C

D

D



# Power Supply for Amplification Circuit



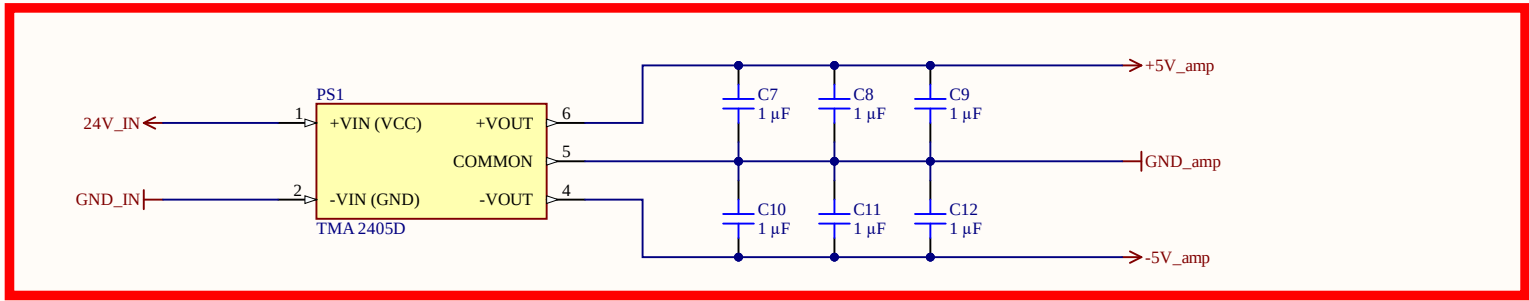
Quelle: DC/DC Wandler 24V auf +/- 5V

Verbraucher: ADA4940 fully differential Amplifier

Ruhestrom: 1,25 mA  
bei 5,0 V  
Power: 0,00625 mW

Output Characteristics:

$U_{out} \sim V_s = 5\text{ V}$   
 $I_{out} = 46\text{ mA peak}$   
 $P_{out max} = 0,23\text{ W}$



Falls USB-C Ladeport realisiert wird -> anderen DC/DC-Wandler auswählend der bis zu 28 V aushält!!

Title Power\_Supply\_Signal\_Conditioning.SchDoc

Revision: Rev01 Design Engineer: Tom Gaupp

Project: uz\_per\_current\_measurement\_dsub\_to\_rj45.PrjPcb

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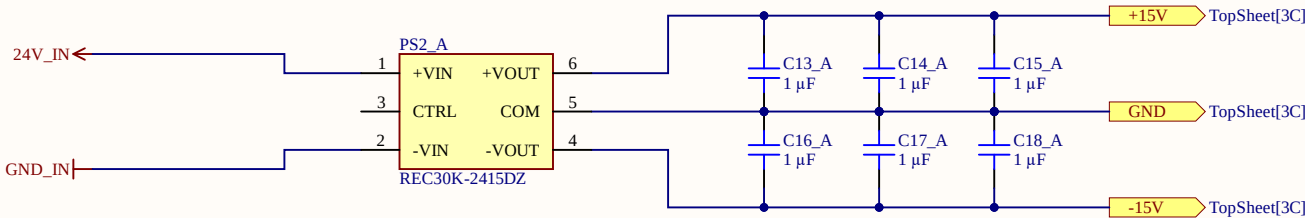
# Power Supply Currentsensors

LEM IN 100-S

Spannungsversorgung: +/- 15 V  
DC Nennstromaufnahmen @15V = 320 mA  
I\_max = 420 mA

Secondary nominal RMS current (@100A) = 0,2 A  
Shielded output cable & shielded plug are recommended to use

Max Power Consumption (pro Stromsensor):  
P = U\*I\_max = 15 V \* 0,420 A = 6,3 W



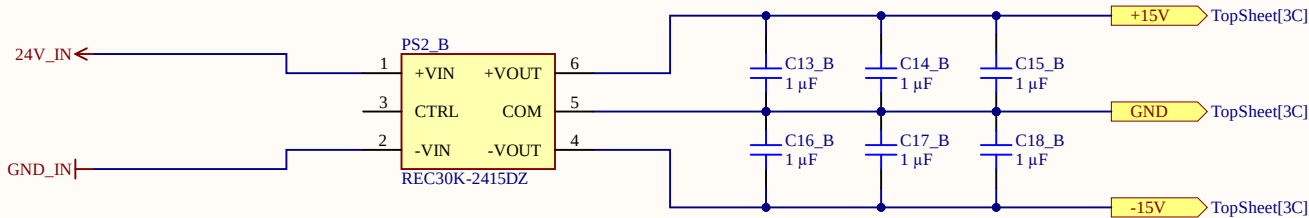
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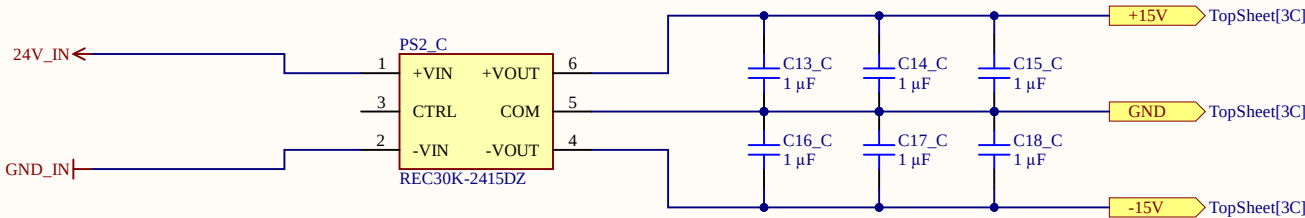
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